

SCOPE OF WORK

RailRover

Prepared by: Bolder Apps | For: Greg Hillabrand | June 2026

1. Problem Statement

Drivers, particularly those running commercial and fleet routes, regularly lose time stuck at railroad crossings blocked by stopped or slow-moving trains. They have no way to know a crossing is blocked until they are already waiting at it, because existing navigation apps show where crossings are but never whether one is blocked right now.

2. Solution Statement

RailRover is a mobile app that shows railroad crossings on a live map, lets drivers report a crossing as blocked or clear, checks fresh blocked reports against the driver's active route in real time, and reroutes the driver around a blocked crossing before they reach it. When entering a destination, drivers can compare routes by how many crossings each one crosses. The MVP launches as a local pilot around Toledo, Ohio that proves the value of real-time, crowd-sourced crossing status.

3. Deliverables

- iOS and Android Mobile Application
- Web Admin Panel

4. User Types

User Type	Description
Driver	The end user. Drives with the app during a trip, sees nearby crossings on the map, reports a crossing as blocked or clear, and receives a warning with an alternate route when a fresh blocked report sits ahead on the active route. Guest access is allowed so a driver can start with minimal setup.
Administrator	Internal platform staff who maintain the crossing inventory, review submitted reports for accuracy, manage driver accounts, and track pilot performance. Access is through the Web Admin Panel only.

5. Module Summary

Module	Description
Foundation	Guest access and lightweight accounts, the live-location layer that powers route awareness, the notification baseline, and user settings shared across the app.
Crossing Map	The Bolder-compiled crossing inventory, the live home map with red, yellow, and green crossing markers, and the crossing detail view.
Crowd-Sourced Reporting	The fast blocked and clear report flows that let any driver update a crossing's status in seconds.
Route Conflict and Alerts	Destination and route options ranked by crossing count, the real-time logic that decides whether a fresh blocked crossing sits ahead on the active route, the alert, the automatic reroute, and the active alerts list.
Web Admin Panel	Back-office tools for administrator sign-in, crossing inventory management, report review, broadcast notifications, and pilot analytics.

6. Module Breakdown

Module 1: Foundation

Feature	Guest Access and Account
Description	A driver can start using the app immediately as a guest to keep first-run friction low, and can create a lightweight account to retain their settings and report history.
User Flow	1. Driver opens the app for the first time. 2. Driver chooses to continue as a guest or create an account. 3. A guest is taken straight to the map; a registering driver provides an email and password. 4. System creates the account and an authenticated session, or a guest session tied to the device. 5. On return visits, the driver resumes their session or signs in.
Edge Cases	A guest later creates an account -> System carries the device settings and recent reports into the new account. An incorrect password is entered -> System responds with "Email or password is incorrect." A driver forgets their password -> System sends a reset link to the registered email.
Acceptance Criteria	1. A driver can reach the map and submit reports as a guest without creating an account. 2. A driver can create an account with email and password and sign in on return visits. 3. Passwords are stored using a one-way salted hash and never in plain text. 4. Each report is tied to a device identifier so guest and account reports can be attributed.

Feature	Location Permissions and Live Location
Description	The app requests location permission and tracks the driver's live position during an active trip so it can place the driver on the map, find the nearest crossing for a report, and check crossings ahead on the route. The location layer is built so always-on background tracking can be enabled in a later phase without rework.
User Flow	1. On first use, the app explains why it needs location and requests permission. 2. Driver grants location access for use while the app is active. 3. System tracks the driver position during an active session and centers the map on it. 4. System uses live position to auto-select the nearest crossing for reports and to evaluate crossings ahead on the route.
Edge Cases	A driver denies location permission -> System explains that route alerts and nearest-crossing reporting need location, and offers a path to enable it in settings. Location signal is temporarily lost -> System holds the last known position and resumes when the signal returns.
Acceptance Criteria	1. The app requests location permission with a clear purpose string before any tracking begins. 2. Live position updates while the app is active and drives map centering and nearest-crossing selection. 3. The location layer is structured so always-on background tracking can be enabled in a later phase. 4. Location data handling follows the privacy disclosures presented at permission time.

Feature	Notifications
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Description	The app delivers push notifications and an in-app inbox for relevant activity, including route alerts surfaced during an active trip and broadcast messages from platform staff. Alerting while the app is closed is part of a later phase.
User Flow	1. System requests notification permission after the driver reaches the map. 2. System sends a push notification and an in-app entry on a relevant event. 3. Driver opens the in-app inbox to review read and unread items. 4. Tapping an item opens the related crossing or alert.
Edge Cases	Push permission is denied -> Alerts and messages remain available in the in-app inbox while the app is active. Several notifications arrive together -> System groups them in the inbox.
Acceptance Criteria	1. Push notifications are delivered to opted-in devices on iOS and Android. 2. An in-app inbox lists notifications with read and unread states. 3. Each notification opens its related crossing or alert. 4. Notification permissions can be managed from settings.

Feature	Settings
Description	A short settings screen where the driver manages notification preferences, the auto-reroute preference, and their account, including signing in or out and deleting the account.
User Flow	1. Driver opens settings. 2. Driver adjusts notification preferences and the auto-reroute preference. 3. Driver signs in, signs out, or deletes their account. 4. System saves the preferences and applies them immediately.
Acceptance Criteria	1. A driver can enable or disable notifications. 2. Auto-reroute is on by default, rerouting around a detected blocked crossing after a short preview; a driver can turn it off to be prompted instead. 3. A driver can sign in, sign out, and delete their account.

Module 2: Crossing Map

Feature	Crossing Inventory
Description	Known railroad crossings for the pilot area are compiled by Bolder Apps from public crossing data, each with its location, road, and city, forming the static set of crossings the app tracks. The pilot covers an approximately 200-mile radius around Toledo, Ohio, and you do not need to supply a crossing list.
User Flow	1. Bolder Apps compiles the pilot crossing set from public data before launch. 2. System stores each crossing with its identifier, name, location, road, and city. 3. System serves crossings to the map and to the report and route logic. 4. Platform staff correct or extend the inventory through the Web Admin Panel.
Edge Cases	A compiled crossing has an inaccurate location -> Platform staff correct its coordinates in the Web Admin Panel so reports and route checks stay accurate.
Acceptance Criteria	1. The pilot crossing set is compiled from public data with identifier, name, location, road, and city for each crossing. 2. Crossings are scoped to the pilot coverage area, an approximately 200-mile radius around Toledo. 3. The inventory is the single source the map, reporting, and route logic read from.

Feature	Home Map and Crossing Markers
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Description	The home screen shows a live map centered on the driver with color-coded markers for nearby crossings, a destination search at the top, report and route actions at the bottom, and a short summary line such as the number of blocked crossings on the active route. A crossing shows red for a fresh blocked report, yellow once a blocked report has aged past the freshness window without a clear report, and green once a clear report is received.
User Flow	1. Driver opens the home map. 2. System centers the map on the driver live location. 3. System displays nearby crossings as red, yellow, or green markers by status. 4. Driver taps a marker to open its detail, or uses the bottom actions to report or enter a destination. 5. System shows a summary line such as "1 blocked crossing on your route" when a route is active.
Edge Cases	A crossing has a fresh blocked report -> System shows it red. A blocked report ages past the freshness window with no clear report -> System shows the crossing yellow, meaning blocked but unconfirmed. A clear report is submitted -> System shows the crossing green; status returns to green only on a clear report, not on a timer.
Acceptance Criteria	1. The map centers on the driver live location. 2. Crossings display as red, yellow, or green markers: red for a fresh blocked report, yellow for a blocked report aged past the freshness window, green once a clear report is received. 3. Destination search is available at the top and report and route actions at the bottom. 4. A summary line reports the count of blocked crossings on the active route.

Feature	Crossing Detail
Description	The crossing detail view shows a crossing's current status, the time of the last report, how many reports it has received, and whether it currently affects the driver's active route.
User Flow	1. Driver taps a crossing marker or list item. 2. System opens the crossing detail. 3. System shows status, last report time, and report count. 4. System indicates whether the crossing affects the active route. 5. Driver can submit a blocked or clear report from the detail view.
Edge Cases	A crossing has no reports yet -> System shows it as unconfirmed with no recent report time.
Acceptance Criteria	1. The detail view shows current status, last report time, and report count. 2. The view indicates whether the crossing is on or near the active route. 3. A driver can report blocked or clear directly from the detail view.

Module 3: Crowd-Sourced Reporting

Feature	Report Blocked
Description	A fast flow that lets a driver mark a crossing as blocked in a couple of taps, in under five seconds, auto-selecting the nearest crossing. Reporting is limited to crossings within about a mile so a driver reports a crossing they can actually see. No note field, to keep it quick while driving.
User Flow	1. Driver taps the blocked report action. 2. System auto-selects the nearest crossing within about a mile of the driver. 3. Driver confirms the crossing.

	4. System saves the report with the crossing, status, timestamp, and device identifier. 5. System updates the crossing to red across the app.
Edge Cases	The driver is not within about a mile of a known crossing -> System does not allow a report, since the driver is too far to observe the crossing. The same crossing is reported blocked repeatedly -> System records the latest report and refreshes the freshness timer.
Acceptance Criteria	1. A blocked report can be submitted in a couple of taps, in under five seconds, with the nearest crossing auto-selected. 2. Reporting is limited to crossings within about a mile of the driver, a configurable distance. 3. Each report saves the crossing, status, timestamp, and device identifier. 4. A new blocked report updates the crossing to red and resets its freshness window. 5. Report submission is rate-limited per device to deter spam.

Feature	Report Clear
Description	The same fast flow used to mark a previously blocked crossing as clear, turning it green. A clear report is the only thing that returns a crossing to green, since the app does not guess a crossing is clear on a timer.
User Flow	1. Driver taps the clear report action. 2. System auto-selects the nearest crossing within about a mile. 3. Driver confirms the crossing. 4. System saves the clear report and updates the crossing to green.
Edge Cases	A crossing is reported clear while another driver reports it blocked -> System keeps the most recent report and reflects it in the crossing status.
Acceptance Criteria	1. A clear report uses the same couple-of-taps flow with the nearest crossing auto-selected. 2. A clear report turns the crossing green and records the time; a crossing returns to green only on a clear report. 3. Reporting is limited to crossings within about a mile of the driver. 4. Clear and blocked reports on the same crossing resolve by most recent report.

Module 4: Route Conflict and Alerts

Feature	Destination Search and Route Options
Description	A driver enters a destination and the app offers route options drawn on the map. Alongside travel time, each option shows how many crossings it passes and how many are currently blocked, so the driver can choose a route that avoids crossings. The driver previews the options before selecting, and the chosen route becomes the active route the alert logic checks against.
User Flow	1. Driver enters a destination in the search field. 2. System geocodes the destination and requests route options. 3. System draws the options on the map and labels each with travel time, total crossings, and currently blocked crossings, with color-coded crossing markers. 4. Driver previews the options and selects one. 5. System sets the chosen route as active for crossing checks.
Edge Cases	A destination cannot be found -> System asks the driver to refine the search. No route is available -> System tells the driver a route could not be generated. Two routes have a similar time but different crossing counts -> System surfaces the crossing counts so the driver can trade a little time for fewer crossings.

Acceptance Criteria	1. A driver can enter a destination and see route options drawn on the map. 2. Each route option shows travel time, the number of crossings on it, and the number currently blocked. 3. A driver can preview the options before selecting one, and the selected route becomes active for the route conflict logic. 4. Route requests handle a not-found destination and a no-route result gracefully.
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Feature	Route Conflict Detection
Description	When a route is active, the app continuously determines whether any crossing with a fresh blocked report sits on the route ahead of the driver, so the driver is warned and rerouted proactively rather than at the last moment. A crossing counts as on the route when it lies within a configurable corridor of the route line, starting at 200 feet. That corridor width decides whether a crossing is on the route; it is not how far ahead the driver is warned.
User Flow	1. System takes the active route line and the driver position. 2. System loads crossings within the route bounding box plus the corridor. 3. System keeps crossings whose distance to the route line is within the corridor, starting at 200 feet. 4. System removes crossings already behind the driver and orders the rest by distance ahead. 5. System checks the remaining crossings for blocked reports still inside the freshness window, starting at 15 minutes. 6. System flags any fresh blocked crossing ahead on the route, however far ahead, and computes an alternate route in the background so a reroute is ready.
Edge Cases	A blocked report ages past the freshness window -> System stops treating the crossing as an active red conflict and shows it yellow. A crossing sits outside the route corridor -> System excludes it so off-route crossings do not trigger alerts. A new blocked report is submitted while a trip is already active -> System updates the active route check in real time, not only from the data present when the trip started.
Acceptance Criteria	1. The 200-foot value is the configurable route corridor width that decides whether a crossing is on the route. 2. A fresh blocked crossing anywhere ahead on the active route is flagged, not only when the driver is near it. 3. Crossings behind the driver are excluded and the rest are ordered by distance ahead. 4. Only blocked reports inside the configurable freshness window, starting at 15 minutes, count as active. 5. New reports submitted during an active trip update the route check in real time.

Feature	Blocked Crossing Alert
Description	When a fresh blocked crossing is found ahead on the route, the app warns the driver with the crossing name or road, the time of the last report, and an estimated time to the crossing when available. With auto-reroute on, the default, the app begins rerouting around the crossing after a short preview countdown the driver can cancel to stay on course; with auto-reroute off, the alert offers reroute, keep current route, and view crossing.
User Flow	1. System detects a fresh blocked crossing ahead on the active route. 2. System presents the alert with the crossing name or road and the last report time. 3. System shows an estimated time to the crossing when it can be calculated. 4. With auto-reroute on, a short preview countdown begins and the app reroutes unless the driver cancels; with it off, the driver chooses reroute, keep current route, or view crossing.

Edge Cases	The crossing is reported clear before the driver reaches it -> System updates or dismisses the alert. Several blocked crossings sit ahead -> System surfaces the nearest one first and lists the rest in active alerts.
Acceptance Criteria	1. An alert fires when a fresh blocked report sits on the route ahead of the driver. 2. The alert shows the crossing name or road and the last report time. 3. With auto-reroute on, the app reroutes after a short preview countdown the driver can cancel; with it off, the alert offers reroute, keep current route, and view crossing. 4. The alert clears when the crossing is reported clear or the driver passes it.

Feature	Reroute
Description	When a blocked crossing is detected ahead, the app generates an alternate route that avoids it, computed in the background so the alternate is ready the moment the driver is alerted. Because avoiding a single point is not a built-in routing capability, the app constructs an avoidance area or alternate waypoints and requests a new route. This is a route-avoidance heuristic on top of the routing service, not turn-by-turn voice navigation. With auto-reroute on, the alternate is applied automatically after a short preview the driver can cancel.
User Flow	1. System computes an alternate route around the blocked crossing in the background. 2. System builds the alternate by constructing an avoidance area or alternate waypoints and re-requesting the route. 3. With auto-reroute on, the app applies the alternate after a short preview countdown; with it off, the driver chooses to accept it. 4. System shows the alternate route and the reason it may be better.
Edge Cases	No reasonable alternate route exists -> System tells the driver no better route was found and keeps the current route. The alternate route passes another blocked crossing -> System evaluates the alternate against the same conflict logic before applying it.
Acceptance Criteria	1. A reroute produces an alternate route that avoids the blocked crossing, computed in the background. 2. With auto-reroute on, the alternate applies automatically after a short preview countdown the driver can cancel; with it off, the driver accepts it. 3. The alternate route is presented with the reason it may be better, and the MVP avoids the single alerted crossing, not multiple crossings at once. 4. The system handles the case where no better alternate exists.

Feature	Active Alerts
Description	A list of the blocked crossings currently relevant to the driver, with crossings on the active route shown first, so the driver can review and act on them in one place.
User Flow	1. Driver opens the active alerts list. 2. System lists fresh blocked crossings relevant to the driver. 3. System orders on-route crossings first. 4. Driver taps a crossing to view its detail or to reroute.
Edge Cases	No crossings are currently relevant -> System shows an empty state.
Acceptance Criteria	1. The list shows fresh blocked crossings relevant to the driver. 2. Crossings on the active route are shown first. 3. Tapping a crossing opens its detail.

Module 5: Web Admin Panel

Feature	Administrator Authentication
Description	Internal platform staff sign in to the Web Admin Panel through a separate authentication flow from drivers, with secure password reset.
User Flow	1. Administrator opens the Web Admin Panel. 2. Administrator enters their credentials. 3. System validates the credentials and opens the dashboard. 4. Administrator can reset their password securely if needed.
Edge Cases	Repeated failed administrator sign-in attempts -> System rate-limits and temporarily locks the account.
Acceptance Criteria	1. Administrator authentication is separate from the driver login. 2. Only accounts with the administrator role can access the panel. 3. Secure password reset is available for administrator accounts.

Feature	Crossing Inventory Management
Description	Internal platform staff add, edit, and remove crossings and correct crossing coordinates so the map, reports, and route checks stay accurate as the pilot area is refined.
User Flow	1. Administrator opens crossing management. 2. Administrator searches for a crossing or adds a new one. 3. Administrator edits a crossing location, road, or city, or removes it. 4. System applies the change across the app.
Edge Cases	An administrator corrects a crossing coordinates -> System updates the marker position and the route checks that depend on it.
Acceptance Criteria	1. Administrators can add, edit, and remove crossings. 2. Coordinate corrections update the map and the route conflict logic. 3. Inventory changes take effect across the app.

Feature	Report Review and User Management
Description	Internal platform staff review submitted blocked and clear reports, remove inaccurate or abusive reports, and manage driver accounts, including suspending a driver who submits false reports.
User Flow	1. Administrator opens the report review queue. 2. System lists recent reports with crossing, status, time, and reporter reference. 3. Administrator removes an inaccurate report or flags a reporter. 4. Administrator searches driver accounts and suspends an abusive reporter where needed.
Edge Cases	A driver submits repeated false reports -> System lets staff suspend the account and discount its reports.
Acceptance Criteria	1. The queue lists recent reports with crossing, status, timestamp, and reporter reference. 2. An administrator can remove an inaccurate report and its effect on crossing state. 3. An administrator can search, suspend, or reinstate driver accounts.

Feature	System-Wide Notifications
Description	Internal platform staff send notifications to all pilot users or a segment for announcements, such as pilot instructions, coverage changes, and maintenance notices.
User Flow	1. Administrator composes a notification.

	- Administrator selects all pilot users or a segment. - Administrator sends the notification. - System delivers it through push and the in-app inbox.
Acceptance Criteria	- An administrator can send a notification to all pilot users or a selected segment. - Notifications are delivered through push and the in-app inbox. - Sent notifications are recorded with their audience and time.

Feature	Pilot Analytics Dashboard
Description	A dashboard showing pilot performance for proof: active and total reports, alert counts, reroutes, and the most affected crossings, over a selected time window.
User Flow	- Administrator opens the analytics dashboard. - System shows active reports, total reports, alert counts, and reroutes. - System lists the most affected crossings. - Administrator changes the time window and the dashboard updates.
Acceptance Criteria	- The dashboard shows active reports, total reports, alert counts, and reroutes. - The dashboard lists the crossings with the most blocked reports. - Metrics update as the selected time window changes.

7. Technical Specifications

Category	Details
Authentication	Guest access plus optional email and password accounts for drivers, with secure password reset, account deletion, and server-managed sessions. Separate authentication for the Web Admin Panel.
Location	Foreground location while the app is active, used for map centering, nearest-crossing reporting, and route checks. The full background-location permission architecture and privacy labels are built in the MVP so always-on background detection can be enabled in a later phase without rework. Location use is disclosed with a clear purpose string at permission time.
Mapping and Routing	Map display, destination geocoding, route options with per-route crossing counts, route polylines, and route distance calculations through the Google Maps SDK and Directions API. Rerouting around a blocked crossing is built by constructing an avoidance area or alternate waypoints, since avoiding a single point is not a built-in routing capability.
Crossing Data	A static crossing inventory compiled by Bolder Apps from public Federal Railroad Administration and related public crossing data, scoped to an approximately 200-mile radius around Toledo, Ohio, with each crossing carrying its identifier, name, latitude, longitude, road, and city.
Crossing Status	Each crossing carries a three-state status: red for a fresh blocked report, yellow once a blocked report ages past the 15-minute freshness window without a clear report, and green only once a clear report is received.
Route Conflict Logic	Bounding-box prefilter, point-to-route-line distance within a configurable corridor that decides whether a crossing is on the route (starting at 200 feet), an ahead-of-driver test, and a configurable freshness window on blocked reports (starting at 15 minutes). A fresh blocked crossing anywhere ahead on the active route is flagged, alternate routes are computed in the background, and the route check updates in real time as new reports arrive during a trip.
Data Records	Crossings (crossing identifier, name, latitude, longitude, road, city) and Reports (report identifier, crossing identifier, status, timestamp, device identifier). Optional route and alert logs support pilot analytics.
Notifications	Push notifications on iOS and Android plus an in-app inbox, delivered server-side to opted-in devices.

Analytics	Pilot metrics for reports, alerts, reroutes, and most-affected crossings, surfaced in the Web Admin Panel.
Security	TLS in transit, AES-256 at rest, OWASP Top 10 compliance, per-device rate limiting on report submission, and report submission restricted to within about a mile of a crossing.
Privacy and App Store Compliance	Location-permission architecture, purpose strings, and privacy labels built to the App Store Review Guidelines and Google Play requirements for location use.
iOS	Latest stable iOS at time of App Store publishing.
Android	Latest stable Android API at time of Google Play publishing.
Browser	Web Admin Panel supports the latest stable versions of major modern browsers (Chrome, Safari, Firefox, Edge).

8. Assumptions and Dependencies

Assumptions

- You provide any available branding assets before design begins.
- Bolder Apps will ensure the application is built in compliance with App Store Review Guidelines.
- Third-party API credentials are provided by you where applicable.
- Any additions outside of this agreed-upon scope will be addressed through a separate addendum.
- RailRover is a working name and is not yet confirmed; the final product name is selected during discovery, before the design phase, since it drives the logo and store listing.
- The MVP launches as a local pilot covering an approximately 200-mile radius around Toledo, Ohio, with the crossing inventory compiled by Bolder Apps from public data. Nationwide crossing coverage is a later phase.
- Crossing status in the MVP is crowd-sourced from driver reports; the app does not detect blockages automatically. MVP alerts surface during an active trip while the app is in use, and the full background-location architecture is built in the MVP, but always-on background detection, sensor fusion, computer vision, train-position integration, and train-speed prediction are a later phase.
- The 15-minute freshness window, the 200-foot route corridor, and the roughly one-mile reporting range ship at these defaults and are tuned during the pilot.
- RailRover provides informational guidance and warnings only. It does not control the vehicle, and drivers remain responsible for their own driving decisions and for obeying actual crossing signals and signage.
- You own the Terms of Service, Privacy Policy, and the data-rights language governing collected location and report data, and Bolder Apps integrates acceptance into the app.

Dependencies

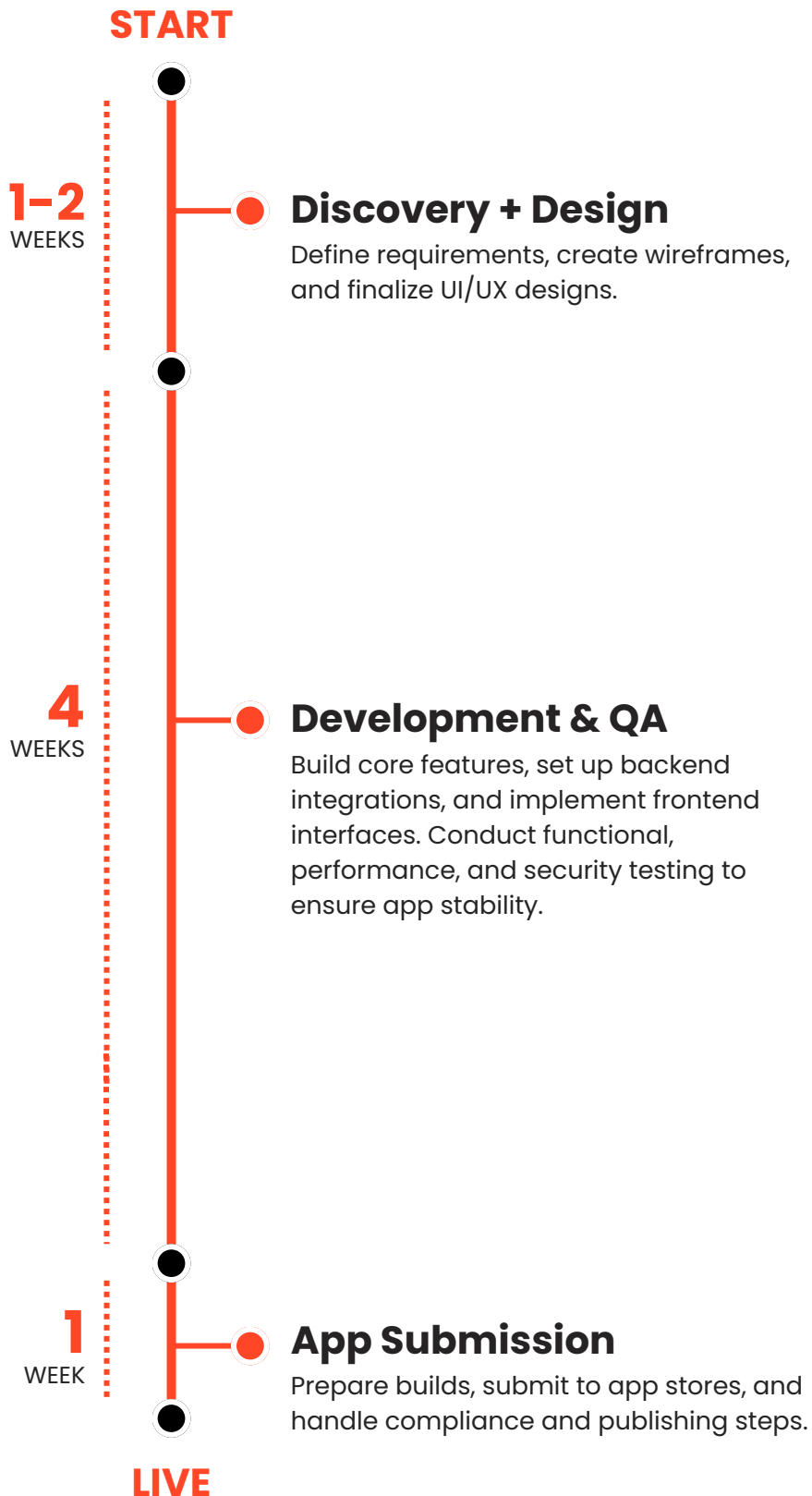
- Apple Developer Program and Google Play Console accounts owned and provided by you.
- You own and configure the domain and DNS records for the Web Admin Panel before launch.
- Patent and freedom-to-operate clearance is confirmed by your counsel before development begins.
- A Google Maps Platform account with billing enabled is provided under your ownership before related development begins.
- You provide finalized brand guidelines, including logo, colors, and typography, before Figma design begins.
- Figma design approval required before development begins.

TECHNOLOGY STACK

Component	Technology	Purpose
Mobile Apps	React Native	Cross-platform iOS and Android app from one codebase, latest stable OS at publishing
Web Admin Panel	React	Browser panel for staff to manage crossings, review reports, and read pilot analytics
Backend Platform	Supabase	Managed Postgres database, authentication, storage, and backend APIs
Maps and Routing	Google Maps Platform	Map display, geocoding, route options with crossing counts, and rerouting
Real-Time	Supabase Realtime	Live crossing-status updates pushed to active trips as new reports arrive
Push Notifications	Expo + FCM	Route alerts and broadcast messages to iOS and Android
Error Monitoring	Sentry	Crash and error monitoring across mobile and web (optional)
Admin Hosting	Vercel	Hosting for the Web Admin Panel
Distribution	App Store + Google Play	Publishing and updates for the apps

Note: *This tech stack is recommended for scalability, performance, and rapid development. We are open to discussing any preferred technologies or alternative solutions to align with your vision.*

TIMELINE



6-7 WEEKS